

Precision

Defined as the measure of the accuracy

Precision measure how many of the positive predictions made by the model were actually correct

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positive} + \text{False Positive}}$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{12}{12+1} = 0,91 = 91\%$$

When precision is important

Precision is important when false predictions are costly
for example

- fraud
- Spam emails

A model with high precision makes fewer incorrect positive predictions

Simple

- Use precision when you want positive predictions to be trustworthy
- Use accuracy when you care about the overall correctness and the dataset is balanced

However, for imbalanced datasets, accuracy can be misleading, so also check precision, recall, and confusion matrix and f1-score.

Recall

is also called the "True Positive rate"